Week 01_Module 01_Part 03

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Variance:

Variance measures the *average squared deviation* of data points from the mean.

The formula for **population variance** is:

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

For a **sample** (a subset of data), we divide by n—1 instead of n:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Difference between Population Variance and Sample Variance

- **Population Variance (σ^2)** measures how spread out the *entire population's* data is from the mean.
- **Sample Variance (s^2)** estimates population variance using a *subset (sample)* of the data.

◇ **Example**

Suppose we have data on exam scores:

[70, 80, 90]

Step 1: Mean

$\bar{x}=(70+80+90)/3=80$

Step 2: Differences from mean

70 → $(70 - 80)^2 = 100$

80 → $(80 - 80)^2 = 0$

90 → $(90 - 80)^2 = 100$

Sum of squared differences = 200

Step 3: Variances

- **Population Variance (σ^2)** = 200 / 3 = **66.67**
- **Sample Variance (s^2)** = 200 / (3 — 1) = **100**

→ The **sample variance is slightly higher** because we divide by a smaller number to correct for bias.

Standard Deviation:

Standard deviation is the square root of variance, giving us a spread measure in the *same unit* as the data:

Example: If the average house price is $200 K with SD = $50 K, most houses lie within $150 K–$250 K.

$$\sigma = \sqrt{\sigma^2}, \quad s = \sqrt{s^2}$$

Variance & SD in ML Pipelines

Variance and standard deviation appear throughout ML:

• **Feature Scaling:** Standardization uses to make features comparable by ensuring that each feature has mean = 0 and standard deviation = 1.

$$z_i = \frac{x_i - \mu}{\sigma}$$

• **Regularization:** Ridge and Lasso reduce coefficient variance to prevent overfitting.

• **Model Diagnostics:** High variance in model performance across folds suggests instability.

Bias-Variance Trade-off

In model evaluation, **variance** also refers to how much a model's predictions change with different training data.

A high-variance model memorizes the training set (overfitting), while low variance but high bias underfits.

So, variance of data and variance of model parameters are different concepts but share the same intuition — *instability due to spread.*

Comparison Table

Concept	**Variance**	**Standard Deviation**
Definition	Average of squared deviations	Square root of variance
Formula	$\Sigma(x-\mu)^2 / n$ or $\Sigma(x-\bar{x})^2 / (n-1)$	$\sqrt{\text{Variance}}$
Unit	Squared unit (e.g., $\$^2$)	Same as data (e.g., \$)
Interpretation	Mathematical measure	Intuitive spread
Use	Theoretical/statistical analysis	Communication & scaling

Both quantify variability, but standard deviation is easier to interpret because it's in the same scale as the data.

Example (same data [70, 80, 90])

We already found variance = 66.67 (population).

Then

Standard Deviation (σ)=66.67=8.16

☑ Interpretation:

- Variance = 66.67 (squared units, less intuitive).
- Standard deviation = 8.16 → means scores typically vary ±8 points from the mean.

Summary & Takeaways

☑ Variance measures *how far* data points deviate from the mean on average.

☑ Standard deviation is the square root of variance, in original units.

☑ Both are critical for understanding data distribution, detecting outliers, and standardizing features.

☑ In ML, low variance features may add little value, while extremely high variance may require scaling or transformation.